Sepehr Mohammadkhani

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Fifth-year PhD candidate in Mathematics specializing in **partial differential equations**, **fluid dynamics**, and **stability analysis**. Current research focuses on **magnetic relaxation** and the **stability of Incompressible porous media (IPM) equations**.

Education

Ph.D. in Mathematics, University of Maryland | College Park, MD | May 2027

Advisor: Prof. Huy Q. Nguyen

B.Sc. in Computer Science, Sharif University of Technology | Tehran, Iran | Feb 2022 | GPA: 3.73 / 4

Minor in Mathematics

Awards

- Hauptman Fellowship, University of Maryland (2025, 2026)

Preprints

- Abdolhamid Banihashemi, **Sepehr Mohammadkhani**, Huy Q. Nguyen. "Sharp stability and instability of stratified steady states for the incompressible porous media equation." Submitted (2026). arXiv:2609.37397 [math.AP].
- **Sepehr Mohammadkhani**, Huy Q. Nguyen. "On Moffatt's magnetic relaxation for 2D and 2.5D flows." submitted (2025). arXiv:2511.15501 [math.AP].

Talks

"**On Moffatt's magnetic relaxation for 2D and 2.5D flows**"

- *Special Session on Stability and instability in nonlinear evolution equations*, Joint Mathematics Meetings (JMM), Washington, D.C. (January 2026)
- Current Trends in Fluid Dynamics, Brin Center, University of Maryland, College Park, MD (April 2026)

"**Sharp stability of the 2D IPM equation**"

- RIT on Applied PDE, University of Maryland, College Park, MD (November 2025)

"**Grenier's iterative scheme for nonlinear instability**"

- RIT on Applied PDE, University of Maryland, College Park, MD (April 2025)

Teaching Experience

Instructor | University of Maryland | College Park, MD | Summers 2023 and 2024

- Math 241 Calculus III: Designed the syllabus, taught all lectures, and wrote all exams.

Teaching Assistant | University of Maryland | College Park, MD | Aug 2022 – Present

- Discussion Sections: Math 113 College Algebra, Math 241 Calculus III (×2), Math 246 Differential Equations for Engineers (×2)
- Grader: Math 410 Advanced Calculus I, Math 432 Introduction to Topology, Stat 410 Introduction to Probability Theory